

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EMT 436 — Past Questions & Materials
by Renance

Course: EMT 436
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

This study pack was compiled and restructured by Renance for EMT 436 students at Federal University of Petroleum Resources, Effurun. It brings together 1 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app
Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 1 — EMT 436 Environmental and Petroleum microbiology

Level 400 • First Semester • PastQuestion

This source paper is a scanned document; its pages are reproduced below.

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF ENVIRONMENTAL MANAGEMENT AND TOXICOLOGY
2024/2025 FIRST SEMESTER EXAMINATION
COURSE CODE/TITLE: EMT436/ENVIRONMENTAL AND PETROLEUM
MICROBIOLOGY

TIME: 2HR 30MINS
DATE: 10TH MAY, 2025
INSTRUCTION: ANSWER FIVE QUESTIONS (ANSWER ALL QUESTIONS IN SECTION A AND ONE QUESTION EACH FROM SECTIONS B, C, D AND E)

SECTION A (COMPULSORY)

- 1a. Discuss how a named body of water can purify itself.
- *b. Explain the sources of pollution in a named environment.
- c. What are the consequences of discharging an effluent rich in nitrogen and phosphorus into an aquatic environment.
- *d. List the significance of microorganisms in the air.
- *e. Enumerate five air-borne infections and their corresponding causative agents.

SECTION B (ANSWER ONE QUESTION)

- 1a. Write on the concept of Environmental Microbiology
- b. Briefly discuss the following:
 - ✓ i. Microbial population dynamics,
 - ✓ ii. Population growth Models
 - ✓ iii. Carrying capacity
 - ✓ iv. Microbial succession
 - ✓ v. Microbial community dynamics
- c. Discuss the various methods of investigating soil micro flora
- 2a. Soil microbiology is the study of microorganisms in soil, their functions, and how they affect soil properties, discuss this with respect to:
 - i. Bacteria (Nitrogen fixation and Biochemical processes)
 - ii. Actinomycetes *actinobacteria, mycorrhizae, nocardia, streptomyces*
 - iii. Fungi
 - iv. Protozoa
 - v. Algae
- b. Itemise the key factors influencing population dynamics.

SECTION C (ANSWER ONE QUESTION)

- 1a. State the rationale for treating sewage effluent before disposal.
- b. What are indicator organisms? State the reasons for their use as indicators of fecal pollution *E. coli*
- c. State the health implication of untreated sewage effluent contamination of drinking water. *cholera*
- d. List the methods that could be used to enumerate microbial load of sewage effluent.
- 2a. A river water source contaminated with sewage effluent was sampled. 1ml of 10^{-4} dilution was plated on MacConkey agar and the culture plates were incubated at 37°C. 250 colonies were counted after 24 hours incubation period. Calculate the coliform load of the water sample in Colony forming units per millilitre (CFU/ml).
- b. Describe the process of conventional treatment of sewage effluent. — *primary, secondary, tertiary*